

01 GET /alarms

RULES ["compatibility aliases derive only from canonical fields", "source R03 requires alarm_type OOS, rule_code R03_CONSEC, and value null", "source SUMMARY requires rule_code SUMMARY_OOC", "source TRACE requires rule_code TRACE_OOS"]

```
{
  "request": {
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    "header": {},
    "path": {},
    "query": {
      "area": {
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        "required": false,
        "schema": {
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          "type": "string"
        },
        "chamber": {
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          "required": false,
          "schema": {
            "type": "string"
          },
          "date_from": {
            "nullable": true,
            "required": false,
            "schema": {
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              "type": "string"
            },
            "date_to": {
            "nullable": true,
            "required": false,
            "schema": {
              "format": "date",
              "type": "string"
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            "nullable": true,
            "required": false,
            "schema": {
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            },
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            "nullable": false,
            "required": false,
            "schema": {
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              "type": "boolean"
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            "parameter": {
            "nullable": true,
            "required": false,
            "schema": {
              "type": "string"
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            "source": {
            "nullable": true,
            "required": true,
            "schema": {
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              "type": "string"
            },
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                    "fields": {
                      "action_code": {
                        "nullable": true,
                        "required": false,
                        "schema": {
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                          "type": "string"
                        },
                      "alarm_id": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "alarm_type": {
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                        "required": true,
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                          "type": "string"
                        },
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                        "required": true,
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                          "type": "string"
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                          "type": "string"
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                        "nullable": true,
                        "required": false,
                        "schema": {
                          "enum": ["FOC", "MFD", "OTH", "RFM", "TMD"],
                          "type": "string"
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                        "nullable": true,
                        "required": false,
                        "schema": {
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                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
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                        "schema": {
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                          "type": "string"
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                        "required": true,
                        "schema": {
                          "type": "string"
                        },
                      "mes_status": {
                        "nullable": true,
                        "required": false,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
                        "schema": {
                          "type": "boolean"
                        },
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                        "required": false,
                        "schema": {
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                        },
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                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
                        "schema": {
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                          "type": "string"
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                        "schema": {
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                        "required": true,
                        "schema": {
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                          "type": "string"
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                      "recipe_step_no": {
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                        "schema": {
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                          "type": "integer"
                        },
                      "rule_code": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "enum": ["R03_CONSEC", "SUMMARY_OOC", "TRACE_OOS"],
                          "type": "string"
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                        "required": false,
                        "schema": {
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                          "type": "integer"
                        },
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                        "required": true,
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                        "schema": {
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                          "type": "integer"
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                        },
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                        "required": false,
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                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
                        "schema": {
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                          "type": "string"
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                                            },
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        }
      }
    }
  }
}
```

02 GET /trace

RULES []

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {
      "chamber": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "lot": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "parameter": {
        "nullable": false,
        "required": true,
        "schema": {
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          "type": "string"
        },
      "wafer": {
        "nullable": false,
        "required": true,
        "schema": {
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          "type": "string"
        }
      }
    },
    "responses": {
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            "fields": {
              "measured_at": {
                "nullable": false,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "recipe_step_no": {
                "nullable": false,
                "required": true,
                "schema": {
                  "minimum": 1.0,
                  "type": "integer"
                },
              "seq_no": {
                "nullable": false,
                "required": true,
                "schema": {
                  "minimum": 0.0,
                  "type": "integer"
                },
              "value": {
                "nullable": false,
                "required": true,
                "schema": {
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                      "required": false,
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                          "fields": {
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                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

03 GET /parameters

RULES []

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {},
    "responses": {
      "200": {
        "schema": {
          "item
```

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s": {"additional_properties": false, "fields": {"LCL": {"nullable": true, "required": false, "schema": {"type": "number"}}, "LSL": {"nullable": true, "required": false, "schema": {"type": "number"}}, "TARGET": {"nullable": false, "required": true, "schema": {"type": "number"}}, "UCL": {"nullable": false, "required": true, "schema": {"type": "number"}}, "USL": {"nullable": false, "required": true, "schema": {"type": "number"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "ctrl_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "ctrl_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "spec_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "spec_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "target_value": {"nullable": false, "required": true, "schema": {"type": "number"}}, "unit": {"nullable": true, "required": false, "schema": {"type": "string"}}, "upper_only": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object", "type": "array", "shape": "array"}}
```

04 POST /documents/search

RULES []

```
{request: {body: {"additional_properties": false, "fields": {"doc_type": {"nullable": true, "required": false, "schema": {"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "query": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "top_k": {"nullable": false, "required": false, "schema": {"default": 4, "maximum": 10.0, "minimum": 1.0, "type": "integer"}}, "type": {"type": "object"}, "header": {"path": {}, "query": {}}, "responses": {"200": {"schema": {"items": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "content": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "score": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": -1.0, "type": "number"}}, "section": {"nullable": true, "required": false, "schema": {"type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"type": "object"}, "array": {"shape": "array"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "array": {"shape": "array"}}, "type": "object"}, "shape": "object"}}
```

05 GET /relations/chambers/{chamber_id}

RULES [{"node_count equals len(nodes)", "relationship_count equals len(relationships)"]

```
{request: {body: null, "header": {}, "path": {"chamber_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "query": {"include_production_context": {"nullable": false, "required": false, "schema": {"default": false, "type": "boolean"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"graph_revision": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "nodes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "display_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "properties": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "type": "object"}, "type": "array"}, "production_context": {"nullable": true, "required": false, "schema": {"additional_properties": false, "fields": {"returned_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "truncated": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}, "relationships": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "target": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}, "root_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}, "type": "object"}, "shape": "object"}}
```

06 GET /agent/runs

RULES []

```
{request: {body: null, "header": {}, "path": {}, "query": {"date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "predicted_fault_code": {"nullable": true, "required": false, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "responses": {"200": {"schema": {"items": {"additional_properties": false, "fields": {"action_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_id"
```

```
: {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_source": {
  "nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}},
"approval_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber": {
  "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {
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```

07 POST /agent/ask

```
RULES [{"fault_code and ground_truth_fault_code are forbidden", "metrology.alarm_result is forbidden"}]
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```

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```

08 GET /approvals

RULES ["APPROVED or REJECTED requires decided_by and decided_at", "PENDING requires all decision fields not null", "only EQP_HOLD actions have approvals"]

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {},
    "responses": {
      "200": {
        "schema": {
          "item s": {
            "additional_properties": false,
            "fields": {
              "action_code": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "EQP_HOLD",
                    "MONITORING",
                    "WARNING"
                  ],
                  "type": "string"
                },
              "action_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "agent_run_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "approval_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "approved_at": {
                "nullable": true,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "approved_by": {
                "nullable": true,
                "required": true,
                "schema": {
                  "type": "string"
                },
              "chamber": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "chamber_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "created_at": {
                "nullable": false,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "decided_at": {
                "nullable": true,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "decided_by": {
                "nullable": true,
                "required": true,
                "schema": {
                  "type": "string"
                },
              "decision_comment": {
                "nullable": true,
                "required": true,
                "schema": {
                  "type": "string"
                },
              "equipment": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "equipment_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "fault_code": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "FOC",
                    "MFD",
                    "OTH",
                    "RFM",
                    "TMD"
                  ],
                  "type": "string"
                },
              "lot": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "lot_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "predicted_fault_code": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "FOC",
                    "MFD",
                    "OTH",
                    "RFM",
                    "TMD"
                  ],
                  "type": "string"
                },
              "reason": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "status": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "APPROVED",
                    "PENDING",
                    "REJECTED"
                  ],
                  "type": "string"
                },
              "type": "object",
              "type": "array",
              "shape": "array"
            }
          }
        }
      }
    }
  }
}
```

09 POST /approvals/{approval_id}/decision

RULES ["APPROVED or REJECTED requires decided_by and decided_at", "PENDING requires all decision fields not null", "only EQP_HOLD actions have approvals"]

```
{
  "request": {
    "body": {
      "additional_properties": false,
      "fields": {
        "decided_by": {
          "nullable": false,
          "required": true,
          "schema": {
            "max_length": 40,
            "min_length": 1,
            "type": "string"
          }
        },
        "decision": {
          "nullable": false,
          "required": true,
          "schema": {
            "enum": ["APPROVED", "REJECTED"],
            "type": "string"
          }
        },
        "decision_comment": {
          "nullable": true,
          "required": false,
          "schema": {
            "max_length": 1000,
            "type": "string"
          }
        },
        "object": {
          "header": {},
          "path": {
            "approval_id": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "string"
              }
            },
            "query": {},
            "responses": {
              "200": {
                "schema": {
                  "additional_properties": false,
                  "fields": {
                    "action_code": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "enum": ["EQP_HOLD", "MONITORING", "WARNING"],
                        "type": "string"
                      }
                    },
                    "action_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "agent_run_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "approval_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "approved_at": {
                      "nullable": true,
                      "required": true,
                      "schema": {
                        "format": "date-time",
                        "type": "string"
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

```
ring"}}, "approved_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"type": "string"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

10 GET /audit-logs

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"actor_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "entity_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "entity_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "event_type": {"nullable": true, "required": false, "schema": {"type": "string"}}}, "responses": {"200": {"schema": {"items": {"additional_properties": true, "fields": {"actor": {"nullable": false, "required": true, "schema": {"type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "detail": {"nullable": true, "required": true, "schema": {"type": "string"}}, "entity": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "type": "object"}, "type": "array"}, "shape": "array"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

11 GET /dataset/bounds

RULES []

deferred inventory

12 GET /dashboard/summary

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"area": {"nullable": true, "required": false, "schema": {"enum": ["ALL", "Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "equipment_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "area": {"nullable": true, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "daily_trend": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"date": {"nullable": false, "required": true, "schema": {"format": "date", "type": "string"}}, "has_r03_consec": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "ooc_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "oos_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "type": "array"}}, "date_range": {"nullable": false, "required": true, "schema": {"items": {"format": "date", "type": "string"}, "type": "array"}}, "equipment_counts": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chambers": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}
```

```
{}}, "hierarchy": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chambers": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "ooc_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "oos_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "pending_approvals": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, 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```

13 GET /alarms/{source}/{alarm_id}

RULES []

```
{ "request": { "body": null, "header": {}, "path": { "alarm_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "query": {}, "responses": { "200": {"schema": {"additional_properties": false, "fields": {"action_code": {"nullable": true, "required": false, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "agent_run_status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": false, "schema": {"enum": ["APPROVED", "AUTO", "EXPIRED", "PENDING", "REJECTED"], "type": "string"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "detail": {"nullable": false, "required": true, "schema": {"type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "hit_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "incident": {"nul
```

```
label": false, "required": true, "schema": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}}, "type": "object"}}, "judgement": {"nullable": false, "required": true, "schema": {"enum": ["OOC", "OOS"], "type": "string"}}, "latest_agent_run_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "lot_hist_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "recipe_step_name": {"nullable": true, "required": false, "schema": {"type": "string"}}, "recipe_step_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "rule_id": {"nullable": false, "required": true, "schema": {"enum": ["R01_OOS", "R02_OOC", "R03_CONSEC"], "type": "string"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "wafer_no": {"nullable": true, "required": false, "schema": {"minimum": 1.0, "type": "integer"}}, "type": "object", "shape": "object", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object", "shape": "object"}}
```

14 GET /alarms/paged

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"area": {"nullable": true, "required": false, "schema": {"enum": ["ALL", "Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "equipment_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "judgement": {"nullable": true, "required": false, "schema": {"enum": ["OOC", "OOS"], "type": "string"}}, "page": {"nullable": false, "required": false, "schema": {"default": 1, "minimum": 1, "type": "integer"}}, "sensor_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "size": {"nullable": false, "required": false, "schema": {"default": 20, "maximum": 100, "minimum": 1, "type": "integer"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"action_code": {"nullable": true, "required": false, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "agent_run_status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": false, "schema": {"enum": ["APPROVED", "AUTO", "EXPIRED", "PENDING", "REJECTED"], "type": "string"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "detail": {"nullable": false, "required": true, "schema": {"type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "hit_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "incident": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_hist_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "recipe_step_name": {"nullable": true, "required": false, "schema": {"type": "string"}}, "recipe_step_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "rule_id": {"nullable": false, "required": true, "schema": {"enum": ["R01_OOS", "R02_OOC", "R03_CONSEC"], "type": "string"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "wafer_no": {"nullable": true, "required": false, "schema": {"minimum": 1.0, "type": "integer"}}, "type": "object", "type": "array"}, "page": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "size": {"nullable": false, "required": true, "schema": {"maximum": 100.0, "minimum": 1.0, "type": "integer"}}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object", "shape": "object", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object", "shape": "object"}}
```

15 GET /traces/catalog

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"areas": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "type": "object", "type": "array"}}, "equipments": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chambe
```

```
rs": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}, "lots": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "wafer_nos": {"nullable": false, "required": true, "schema": {"items": {"type": "integer"}, "type": "array"}}, "type": "object"}, "type": "array"}, "recipes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "recipe_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}, "sensors": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"ctrl_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "ctrl_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "sensor_name": {"nullable": false, "required": true, "schema": {"type": "string"}}, "spec_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "spec_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "target": {"nullable": true, "required": false, "schema": {"type": "number"}}, "unit": {"nullable": true, "required": false, "schema": {"type": "string"}}, "upper_only": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

16 POST /traces/search

RULES []

```
{ "request": { "body": { "additional_properties": false, "fields": { "area": { "nullable": true, "required": false, "schema": { "enum": ["Etch", "Photo"], "type": "string" } }, "chamber_id": { "nullable": true, "required": false, "schema": { "min_length": 1, "type": "string" } }, "equipment_id": { "nullable": true, "required": false, "schema": { "min_length": 1, "type": "string" } }, "from": { "nullable": true, "required": false, "schema": { "format": "date-time", "type": "string" } }, "lot_id": { "nullable": true, "required": false, "schema": { "min_length": 1, "type": "string" } }, "recipe_id": { "nullable": true, "required": false, "schema": { "min_length": 1, "type": "string" } }, "sensor_ids": { "nullable": true, "required": false, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, "to": { "nullable": true, "required": false, "schema": {"format": "date-time", "type": "string"}}, "wafer_nos": {"nullable": true, "required": false, "schema": {"items": {"type": "integer"}, "type": "array"}}, "type": "object"}, "header": {}, "path": {}, "query": {}, "responses": { "200": { "schema": { "additional_properties": false, "fields": {"limits": {"nullable": false, "required": true, "schema": {"additional_properties": {"additional_properties": false, "fields": {"ctrl_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "ctrl_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "sensor_name": {"nullable": false, "required": true, "schema": {"type": "string"}}, "spec_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "spec_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "target": {"nullable": true, "required": false, "schema": {"type": "number"}}, "unit": {"nullable": true, "required": false, "schema": {"type": "string"}}, "upper_only": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}, "fields": {}, "type": "object"}}, "measured_step_stats": {"nullable": false, "required": false, "schema": {"additional_properties": {"additional_properties": true, "fields": {}, "type": "object"}, "fields": {}, "type": "object"}}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "wafers": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_hist_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "missing_steps": {"nullable": false, "required": false, "schema": {"items": {"type": "string"}, "type": "array"}}, "occurred_at": {"nullable": true, "required": false, "schema": {"format": "date-time", "type": "string"}}, "points": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"measured_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "recipe_step_name": {"nullable": true, "required": false, "schema": {"type": "string"}}, "recipe_step_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "seq_no": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "value": {"nullable": false, "required": true, "schema": {"type": "number"}}, "type": "object"}, "type": "array"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "wafer_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}, "422": { "schema": { "additional_properties": true, "fields": { "detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "array"}}, "type": "object"}, "shape": "object" } }
```

17 GET /relations/equipment/{equipment_id}

RULES []

deferred inventory

18 GET /documents/{document_id}

RULES []

```
{ "request": { "body": null, "header": {}, "path": { "document_id": {"nullable": false, "required": true, "schema": {"type": "string"} } }, "query": {}, "responses": { "200": { "schema": { "additional_properties": false, "fields": {"chunks": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string" } } } }
```

```
{ "string": "chunk_seq", "nullable": false, "required": true, "schema": { "minimum": 0.0, "type": "integer" }, "content": { "nullable": false, "required": true, "schema": { "type": "string" } }, "section_title": { "nullable": true, "required": false, "schema": { "type": "string" } }, "type": "object", "type": "array" }, { "doc_type": { "nullable": true, "required": false, "schema": { "enum": [ "MANUAL", "SPEC", "TROUBLESHOOT" ], "type": "string" } }, "document_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "model_code": { "nullable": true, "required": false, "schema": { "min_length": 1, "type": "string" } }, "source_path": { "nullable": true, "required": false, "schema": { "type": "string" } }, "title": { "nullable": false, "required": true, "schema": { "type": "string" } }, "version": { "nullable": true, "required": false, "schema": { "type": "string" } }, "type": "object", "shape": "object", "422": { "schema": { "additional_properties": true, "fields": { "detail": { "nullable": false, "required": false, "schema": { "items": { "additional_properties": true, "fields": { "loc": { "nullable": false, "required": true, "schema": { "items": { "type": "union", "variants": { { "type": "string", "type": "integer" } }, "type": "array" } }, "msg": { "nullable": false, "required": true, "schema": { "type": "string" } }, "type": { "nullable": false, "required": true, "schema": { "type": "string" } } }, "type": "object", "type": "array" } }, "type": "object", "shape": "object" } } }
```

```
19 POST /agent/runs
```

```
RULES []
```

```
{ "request": { "body": { "additional_properties": false, "fields": { "alarm": { "nullable": false, "required": true, "schema": { "additional_properties": false, "fields": { "alarm_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "source": { "nullable": false, "required": true, "schema": { "enum": [ "R03", "SUMMARY", "TRACE" ], "type": "string" } }, "type": "object" } }, "header": { }, "path": { }, "query": { }, "responses": { "202": { "schema": { "additional_properties": false, "fields": { "agent_run_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "alarm": { "nullable": false, "required": true, "schema": { "additional_properties": false, "fields": { "alarm_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "source": { "nullable": false, "required": true, "schema": { "enum": [ "R03", "SUMMARY", "TRACE" ], "type": "string" } }, "status": { "nullable": false, "required": true, "schema": { "type": "string" } }, "type": "object", "shape": "object", "422": { "schema": { "additional_properties": true, "fields": { "detail": { "nullable": false, "required": false, "schema": { "items": { "additional_properties": true, "fields": { "loc": { "nullable": false, "required": true, "schema": { "items": { "type": "union", "variants": { { "type": "string", "type": "integer" } }, "type": "array" } }, "msg": { "nullable": false, "required": true, "schema": { "type": "string" } }, "type": { "nullable": false, "required": true, "schema": { "type": "string" } } }, "type": "object", "type": "array" } }, "type": "object", "shape": "object" } } } }
```

```
20 GET /agent/runs/{run_id}
```

```
RULES [delivery_policy_lub204lub77dubc740 ACTION-POLICY-V1 lud638lub658lub774luba70 MOCK-NOTIFY-V1lub740lub2b9lubc778lubc18lub824lub00b7lub655lubc778 lubae30lub85d lubc5c6lubc774 lubc774luba54lubc77c lubc0f lubc790lub3d9 Kafka/MES MockLubc744 lubc5f0lub3d9lubd55club2e4; luc2b9lubc778 lubc0c1ud0d0club294 nulllubc774lub2e4]
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[illegible]


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22 POST /agent/runs/{run_id}/retry
RULES []
deferred inventory
23 GET /agent/runs/paged
RULES []
deferred inventory
24 GET /approvals/paged
RULES []
deferred inventory
25 GET /actions

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26 GET /actions/{action_id}
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EGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND",
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RNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOUR
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}}}
27 POST /actions/{action_id}/deliveries/{channel}/retry
RULES []
deferred inventory
28 POST /internal/actions/{action_id}/delivery
RULES [{"FAILED requires error_code non-null and permits provider_message_id null", "SENT requires provide
r_message_id non-null and error_code null"}]
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29 POST /analytics/query
RULES []
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30 POST /analytics/graph-query

RULES []

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31 POST /analytics/validate

RULES []

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32 GET /analytics/history

RULES []

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33 GET /analytics/evaluations
RULES []
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34 GET /audit-logs/paged
RULES []
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"schema": {"type": "string"}}, "event": {"nullable": false, "required": true,

```

[illegible]

```
e, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "rag": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "reference_migration": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"}}, "type": "object"}}, "type": "object"}}, "dataset_epoch": {"nullable": false, "required": true, "schema": {"type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["NOT_READY", "READY"], "type": "string"}}, "type": "object"}, "shape": "object"}}}
```

DTO COMPONENTS

ActionCode

```
{"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}
```

ActionDeliveryDetailItem

```
{"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "completed_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "started_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}
```

ActionDeliveryItem

```
{"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}
```

ActionDetailResponse

```
{"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "created_by_agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "completed_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "started_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}, "type": "array"}}, "delivery_policy": {"nullable": false, "required": false, "schema": {"default": "ACTION-POLICY-V1", "enum": ["ACTION-POLICY-V1", "MOCK-NOTIFY-V1"], "type": "string"}}, "equipment": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
```

ActionItem

```
{"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "created_by_agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}, "type": "array"}}, "delivery_policy": {"nullable": false, "required": false, "schema": {"default": "ACTION-POLICY-V1", "enum": ["ACTION-POLICY-V1", "MOCK-NOTIFY-V1"], "type": "string"}}, "equipment": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}
```

```
{}}, "type": "object")
```

AgentAskCompatibilityEvidence

```
{
  "additional_properties": false,
  "fields": {
    "chunk_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "doc_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "document_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "section": {
      "nullable": true,
      "required": true,
      "schema": {
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

AgentAskRequest

```
{
  "additional_properties": false,
  "fields": {
    "agent_run_id": {
      "nullable": true,
      "required": false,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "question": {
      "nullable": false,
      "required": true,
      "schema": {
        "max_length": 1000,
        "min_length": 1,
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

AgentAskResponse

```
{
  "additional_properties": false,
  "fields": {
    "answer": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "string"
      }
    },
    "confidence": {
      "nullable": true,
      "required": true,
      "schema": {
        "maximum": 1,
        "minimum": 0,
        "type": "number"
      }
    },
    "evidence": {
      "nullable": true,
      "required": true,
      "schema": {
        "additional_properties": false,
        "fields": {
          "chunk_id": {
            "nullable": false,
            "required": true,
            "schema": {
              "min_length": 1,
              "type": "string"
            }
          },
          "doc_id": {
            "nullable": false,
            "required": true,
            "schema": {
              "min_length": 1,
              "type": "string"
            }
          },
          "document_id": {
            "nullable": false,
            "required": true,
            "schema": {
              "min_length": 1,
              "type": "string"
            }
          },
          "section": {
            "nullable": true,
            "required": true,
            "schema": {
              "type": "string"
            }
          }
        },
        "type": "object"
      }
    },
    "evidence_items": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "discriminator": "type",
          "type": "discriminated_union",
          "variants": {
            "ALARM": {
              "additional_properties": false,
              "fields": {
                "excerpt": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  }
                },
                "required": true,
                "schema": {
                  "type": "string"
                }
              },
              "source_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                }
              },
              "title": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                }
              },
              "type": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": ["ALARM"],
                  "type": "string"
                }
              }
            },
            "DOCUMENT": {
              "additional_properties": false,
              "fields": {
                "chunk_id": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  }
                },
                "document_id": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  }
                },
                "excerpt": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "type": "string"
                  }
                },
                "section": {
                  "nullable": true,
                  "required": true,
                  "schema": {
                    "type": "string"
                  }
                },
                "source_id": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  }
                },
                "title": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  }
                },
                "type": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "enum": ["DOCUMENT"],
                    "type": "string"
                  }
                }
              },
              "type": "object"
            },
            "GRAPH": {
              "additional_properties": false,
              "fields": {
                "excerpt": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "type": "string"
                  }
                },
                "graph_revision": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "type": "string"
                  }
                },
                "relation_id": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "pattern": "^\\[0-9a-f\\{64\\}$",
                    "type": "string"
                  }
                },
                "source_id": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  }
                },
                "title": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  }
                },
                "type": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "enum": ["GRAPH"],
                    "type": "string"
                  }
                }
              },
              "type": "object"
            },
            "METROLOGY": {
              "additional_properties": false,
              "fields": {
                "excerpt": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "type": "string"
                  }
                },
                "source_id": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  }
                },
                "title": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  }
                },
                "type": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "enum": ["METROLOGY"],
                    "type": "string"
                  }
                }
              },
              "rules": [
                "metrology.alarm_result is forbidden"
              ],
              "type": "object"
            },
            "TRACE": {
              "additional_properties": false,
              "fields": {
                "excerpt": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "type": "string"
                  }
                },
                "source_id": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  }
                },
                "title": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  }
                },
                "type": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "enum": ["TRACE"],
                    "type": "string"
                  }
                }
              },
              "type": "object"
            }
          },
          "type": "array"
        },
        "limit": {
          "nullable": false,
          "required": true,
          "schema": {
            "type": "string"
          }
        },
        "limitations": {
          "nullable": false,
          "required": true,
          "schema": {
            "items": {
              "type": "string"
            },
            "type": "array"
          }
        },
        "predicted_fault_code": {
          "nullable": true,
          "required": true,
          "schema": {
            "enum": ["FOC", "MFD", "OTH", "RFM", "TMD"],
            "type": "string"
          }
        },
        "recommended_action": {
          "nullable": true,
          "required": true,
          "schema": {
            "enum": ["EQP_HOLD", "MONITORING", "WARNING"],
            "type": "string"
          },
          "title": {
            "nullable": false,
            "required": true,
            "schema": {
              "min_length": 1,
              "type": "string"
            }
          },
          "tools": {
            "nullable": false,
            "required": true,
            "schema": {
              "items": {
                "additional_properties": false,
                "fields": {
                  "name": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "min_length": 1,
                      "type": "string"
                    }
                  },
                  "result": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "min_length": 1,
                      "type": "string"
                    }
                  },
                  "result_summary": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "min_length": 1,
                      "type": "string"
                    }
                  },
                  "status": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "enum": ["ERROR", "SUCCESS", "TIMEOUT"],
                      "type": "string"
                    }
                  },
                  "tool_name": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "min_length": 1,
                      "type": "string"
                    }
                  }
                },
                "type": "object"
              },
              "type": "array"
            },
            "rules": [
              "fault_code and ground_truth_fault_code are forbidden"
            ],
            "type": "object"
          }
        }
      }
    }
  },
  "type": "object"
}
```

AgentEvaluationResponse

```
{
  "additional_properties": false,
  "fields": {
    "fault_5class": {
      "nullable": true,
      "required": true,
      "schema": {
        "additional_properties": false,
        "fields": {
          "classification": {
            "nullable": false,
            "required": true,
            "schema": {
              "additional_properties": false,
              "fields": {
                "accuracy": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "additional_properties": false,
                    "fields": {
                      "denominator": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "minimum": 0.0,
                          "type": "integer"
                        }
                      },
                      "numerator": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "minimum": 0.0,
                          "type": "integer"
                        }
                      },
                      "rate": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "maximum": 1.0,
                          "minimum": 0.0,
                          "type": "number"
                        }
                      }
                    },
                    "type": "object"
                  },
                  "by_class": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "additional_properties": {
                        "additional_properties": false,
                        "fields": {
                          "f1": {
                            "nullable": false,
                            "required": true,
                            "schema": {
                              "maximum": 1.0,
                              "minimum": 0.0,
                              "type": "number"
                            }
                          },
                          "false_negative": {
                            "nullable": false,
                            "required": true,
                            "schema": {
                              "minimum": 0.0,
                              "type": "integer"
                            }
                          },
                          "false_positive": {
                            "nullable": false,
                            "required": true,
                            "schema": {
                              "minimum": 0.0,
                              "type": "integer"
                            }
                          },
                          "precision": {
                            "nullable": false,
                            "required": true,
                            "schema": {
                              "maximum": 1.0,
                              "minimum": 0.0,
                              "type": "number"
                            }
                          },
                          "recall": {
                            "nullable": false,
                            "required": true,
                            "schema": {
                              "maximum": 1.0,
                              "minimum": 0.0,
                              "type": "number"
                            }
                          }
                        },
                        "type": "object"
                      }
                    }
                  }
                }
              },
              "type": "object"
            }
          }
        },
        "type": "object"
      }
    }
  },
  "type": "object"
}
```



```
fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}},
"source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type":
"string"}}}, {"type": "object"}, {"type": "array"}}, {"supporting_chunk_ids": {"nullable": false, "required
": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, {"supporting_relation
_ids": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "ty
pe": "array"}}, {"uncertainty": {"nullable": false, "required": true, "schema": {"max_length": 1000, "type
": "string"}}}, {"type": "object"}
```

AgentRunAccepted

```
{"additional_properties": false, "fields": {"agent_run_id": {"nullable": false, "required": true, "schema
": {"min_length": 1, "type": "string"}}, "alarm": {"nullable": false, "required": true, "schema": {"addit
ional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_le
ngth": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03",
"SUMMARY", "TRACE"], "type": "string"}}}, {"type": "object"}}, {"status": {"nullable": false, "required": t
rue, "schema": {"enum": ["RUNNING"], "type": "string"}}, {"type": "object"}
```

AgentRunAcceptedResponse

```
{"additional_properties": false, "fields": {"agent_run_id": {"nullable": false, "required": true, "schema
": {"min_length": 1, "type": "string"}}, "alarm": {"nullable": false, "required": true, "schema": {"addit
ional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_le
ngth": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03",
"SUMMARY", "TRACE"], "type": "string"}}}, {"type": "object"}}, {"status": {"nullable": false, "required": t
rue, "schema": {"type": "string"}}, {"type": "object"}
```

AgentRunActionItem

```
{"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema"
: {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, {"action_id": {"nullable": false, "r
equired": true, "schema": {"min_length": 1, "type": "string"}}, {"agent_run_id": {"nullable": false, "requ
ired": true, "schema": {"min_length": 1, "type": "string"}}, {"approval_status": {"nullable": true, "requi
red": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, {"deliveries": {"n
ullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"chann
el": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, {"comp
leted_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, {"st
arted_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, {"st
atus": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDIN
G", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}}, {"type": "object"}, {"type": "array"}}, {"delivery_p
olicy": {"nullable": false, "required": false, "schema": {"default": "ACTION-POLICY-V1", "enum": ["ACTION
-POLICY-V1", "MOCK-NOTIFY-V1"], "type": "string"}}, {"reason": {"nullable": false, "required": true, "sche
ma": {"min_length": 1, "type": "string"}}, {"type": "object"}
```

AgentRunApprovalItem

```
{"additional_properties": false, "fields": {"action_id": {"nullable": false, "required": true, "schema":
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AlarmEvidence
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AlarmItem
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AlarmSource
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AnalysisQueryRequest
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AnalysisQueryResponse
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AuditLogPageResponse

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AutoToolCallItem

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ChamberGraphResponse

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ChamberRelationResponse

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ChatToolCallItem
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DeliveryResult
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DocumentAskEvidence

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DocumentChunkItem

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DocumentDetailResponse

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DocumentEvidence

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DocumentHit

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DocumentSearchRequest

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DocumentType

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ErrorCode

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ErrorResponse

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EvaluationListResponse

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EvidenceAssessmentBlock

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Evidenceltem

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FaultHypothesis
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[illegible]

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AUDIT EVENTS
DETECTION_COMPLETED / LOT_HIST / A
AGENT_RUN_STARTED / AGENT_RUN / C
HYPOTHESIS_GENERATED / AGENT_RUN / C
APPROVAL_REQUESTED / APPROVAL / C
APPROVAL_DECIDED / APPROVAL / C
ACTION_SENT / ACTION / C delivery service
ACTION_SEND_FAILED / ACTION / C delivery service
AGENT_RUN_COMPLETED / AGENT_RUN / C
AGENT_RUN_FAILED / AGENT_RUN / C
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